

Backend Technical Documentation

Seven-service backend architecture and operations.



This document explains its subject first in plain language, then in engineering detail. It is grounded in the Craves repository snapshot and deliberately separates current implementation, legacy/reference code, milestone foundations and repository gaps so readers do not mistake aspiration for proof.

Version 1.0 | Evidence date: 14 August 2026 | Repository: [rmorampudi09-arch/Craves-Build-platform](#) | Snapshot commit: [3225f7fa531a6b07185fb5e034f7930f8bd8b571](#)

service boundaries - plain-language purpose

Plain-English explanation

Craves includes a governed semantic meal concierge for assisted discovery rather than an authority that can bypass commerce rules.

How it works technically

Recent main history and [apps/customer-web-next/docs/signalr-ai-concierge.md](#) describe SignalR semantic concierge work. Recommendations still rely on governed product data and ordinary catalog, quote, authorization and order paths for transactions.

Flow described in words

1. Actor starts the capability with the appropriate identity/context. 2. Client/BFF validates local prerequisites and sends an API request. 3. APIM and the owning service enforce identity, role, ownership and business-state rules. 4. The domain service reads/writes authoritative state and calls providers only through defined boundaries. 5. The caller receives authoritative result or a traceable error; support uses request/correlation evidence rather than secrets.

Errors, edge cases and controls

- Stale UI/cache must not override server state.
- UI visibility is not authorization evidence.
- Retries of money/state-changing operations require idempotency awareness.
- Provider/configuration outages should preserve core domain consistency where safe.
- Missing repository proof is reported as a gap rather than converted into a production claim.

Diagnostic and validation steps

1. Capture time, environment, operation, expected/actual result and a user-safe identifier. 2. Capture HTTP status, stable error code and request/correlation ID; never copy OTPs, passwords, access tokens or provider secrets. 3. Verify caller role/ownership and current authoritative state before retrying. 4. Check owning service health, then its provider/dependency after local/configuration causes are excluded. 5. If a release is implicated, compare artifact/config marker with prior known-good revision and validate rollback compatibility.

Data, security and deployment impact

The ai boundary owns its authoritative state; clients request or display changes but do not become the source of truth. Secrets and unnecessary personal data stay outside logs and PDFs. Releases must keep API, policy, schema and artifact compatible, with health/readiness evidence and rollback/backout appropriate to the change.

Evidence status

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The active backend target is seven Spring Boot services: auth, user-chef, catalog, order, subscription, integration and notification.

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order-service documents Problem Details fields and codes including ORDER_NOT_FOUND, ORDER_IDEMPOTENCY_CONFLICT, INVALID_STATUS, CANCELLATION_NOT_ALLOWED, REFUND_FAILED, UNAUTHORIZED, FORBIDDEN, BACKEND_CHANGE_DISABLED and BACKEND_UNAVAILABLE. Diagnostics should start from request/correlation evidence.

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service boundaries - deployment, security and operational validation

Plain-English explanation

The repository has broad CI/CD references across backend, frontend, mobile, APIM, data, privacy, security, smoke and release controls.

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customer-web-next README describes ACR build/push, Container App revision deployment, readiness checks and restoration of the prior image/revision on failed readiness. Pipeline names not resolvable at their cited path are classified as reference-only instead of assigned invented triggers.

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Implemented on main. Evidence: apps/customer-web-next/README.md; repository YAML search; docs/handover/. Snapshot: main @ 3225f7fa531a (14 August 2026).

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Secure HTTP-only cookies keep web token state away from ordinary browser JavaScript. Backend services evaluate roles such as USER, CHEF and ADMIN and may also enforce object ownership. Mobile milestone guidance stores refresh credentials in Keychain/Keystore and access tokens in memory.

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Data, security and deployment impact

The errors boundary owns its authoritative state; clients request or display changes but do not become the source of truth. Secrets and unnecessary personal data stay outside logs and PDFs. Releases must keep API, policy, schema and artifact compatible, with health/readiness evidence and rollback/backout appropriate to the change.

Evidence status

Implemented on main. Evidence: services/order-service/README.md; services/notification-service/README.md; apps/customer-web-next/README.md. Snapshot: main @ 3225f7fa531a (14 August 2026).

user profiles - deployment, security and operational validation

Plain-English explanation

The repository has broad CI/CD references across backend, frontend, mobile, APIM, data, privacy, security, smoke and release controls.

How it works technically

customer-web-next README describes ACR build/push, Container App revision deployment, readiness checks and restoration of the prior image/revision on failed readiness. Pipeline names not resolvable at their cited path are classified as reference-only instead of assigned invented triggers.

Flow described in words

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Errors, edge cases and controls

- Stale UI/cache must not override server state.
- UI visibility is not authorization evidence.
- Retries of money/state-changing operations require idempotency awareness.
- Provider/configuration outages should preserve core domain consistency where safe.
- Missing repository proof is reported as a gap rather than converted into a production claim.

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Data, security and deployment impact

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Evidence status

Implemented on main. Evidence: apps/customer-web-next/README.md; repository YAML search; docs/handover/. Snapshot: main @ 3225f7fa531a (14 August 2026).

chef profiles/verification - plain-language purpose

Plain-English explanation

Chef mode is the supply side of the marketplace: onboarding, verification, menus, kitchen/order operations and earnings.

How it works technically

user-chef-service persists chef profiles and verification state. The frontend can switch context, but the server remains the authority: showing Chef mode never grants a CHEF role by itself.

Flow described in words

1. Actor starts the capability with the appropriate identity/context. 2. Client/BFF validates local prerequisites and sends an API request. 3. APIM and the owning service enforce identity, role, ownership and business-state rules. 4. The domain service reads/writes authoritative state and calls providers only through defined boundaries. 5. The caller receives authoritative result or a traceable error; support uses request/correlation evidence rather than secrets.

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Evidence status

Implemented on main. Evidence: services/user-chef-service/README.md; apps/customer-web-next/src/components/chef/; apps/customer-web-next/src/components/chef-mode/. Snapshot: main @ 3225f7fa531a (14 August 2026).

chef profiles/verification - technical architecture

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Evidence status

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catalog-service - plain-language purpose

Plain-English explanation

Catalog is the source of browseable meal and cuisine information used by discovery and commerce validation.

How it works technically

catalog-service is a Spring Boot/Flyway service with JWT controls. The web contains discovery, meals, search and selector modules. Administrative catalog operations are described by APIM guidance, including bulk pause/import and SHA-256 integrity checks.

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Implemented on main. Evidence: services/catalog-service/README.md; apps/customer-web-next/src/components/discovery/; apps/customer-web-next/src/components/meals/; infra/apim/README.md. Snapshot: main @ 3225f7fa531a (14 August 2026).

catalog-service - technical architecture

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catalog-service - end-to-end flow

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order creation/idempotency - plain-language purpose

Plain-English explanation

Order management is the transactional spine of Craves: create, retrieve, status, cancellation/refund, delivery state, ETA/shortage, proof and administrative controls.

How it works technically

order-service documents idempotency, structured Problem Details errors, reconciliation, provider integration and controlled multi-database routing. Supported roles include customer, chef, admin and delivery-partner contexts with ownership enforcement.

Flow described in words

1. Actor starts the capability with the appropriate identity/context. 2. Client/BFF validates local prerequisites and sends an API request. 3. APIM and the owning service enforce identity, role, ownership and business-state rules. 4. The domain service reads/writes authoritative state and calls providers only through defined boundaries. 5. The caller receives authoritative result or a traceable error; support uses request/correlation evidence rather than secrets.

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Evidence status

Implemented on main. Evidence: services/order-service/README.md; apps/customer-web-next/src/components/order-history/; apps/customer-web-next/src/components/order-tracking/. Snapshot: main @ 3225f7fa531a (14 August 2026).

order creation/idempotency - technical architecture

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order-service documents idempotency, structured Problem Details errors, reconciliation, provider integration and controlled multi-database routing. Supported roles include customer, chef, admin and delivery-partner contexts with ownership enforcement.

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order creation/idempotency - deployment, security and operational validation

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order state - plain-language purpose

Plain-English explanation

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order state - technical architecture

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cancellation/refund - plain-language purpose

Plain-English explanation

Payments are separated from raw card handling so the Craves UI can use provider-hosted entry while the backend protects order/payment state.

How it works technically

The web documents Cashfree hosted payment sessions. order-service verifies Razorpay callbacks with HMAC SHA-256 over the raw body using X-Razorpay-Signature, and integration-service documents normalized payment-link operations.

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delivery/ETA/shortage - plain-language purpose

Plain-English explanation

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- UI visibility is not authorization evidence.
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Data, security and deployment impact

The order boundary owns its authoritative state; clients request or display changes but do not become the source of truth. Secrets and unnecessary personal data stay outside logs and PDFs. Releases must keep API, policy, schema and artifact compatible, with health/readiness evidence and rollback/backout appropriate to the change.

Evidence status

Implemented on main. Evidence: services/order-service/README.md; apps/customer-web-next/src/components/order-history/; apps/customer-web-next/src/components/order-tracking/. Snapshot: main @ 3225f7fa531a (14 August 2026).

delivery/ETA/shortage - errors and edge cases

Plain-English explanation

Order management is the transactional spine of Craves: create, retrieve, status, cancellation/refund, delivery state, ETA/shortage, proof and administrative controls.

How it works technically

order-service documents idempotency, structured Problem Details errors, reconciliation, provider integration and controlled multi-database routing. Supported roles include customer, chef, admin and delivery-partner contexts with ownership enforcement.

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delivery/ETA/shortage - deployment, security and operational validation

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proof/media trust - plain-language purpose

Plain-English explanation

Craves includes a governed semantic meal concierge for assisted discovery rather than an authority that can bypass commerce rules.

How it works technically

Recent main history and apps/customer-web-next/docs/signalr-ai-concierge.md describe SignalR semantic concierge work. Recommendations still rely on governed product data and ordinary catalog, quote, authorization and order paths for transactions.

Flow described in words

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Data, security and deployment impact

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Evidence status

Implemented on main. Evidence: apps/customer-web-next/docs/signalr-ai-concierge.md; commit 71e795b. Snapshot: main @ 3225f7fa531a (14 August 2026).

proof/media trust - technical architecture

Plain-English explanation

The active backend target is seven Spring Boot services: auth, user-chef, catalog, order, subscription, integration and notification.

How it works technically

Services own focused business domains, validate JWT role/ownership, expose health endpoints, evolve schema through Flyway where documented, and isolate provider-specific behavior behind integration/notification boundaries.

Flow described in words

1. Actor starts the capability with the appropriate identity/context. 2. Client/BFF validates local prerequisites and sends an API request. 3. APIM and the owning service enforce identity, role, ownership and business-state rules. 4. The domain service reads/writes authoritative state and calls providers only through defined boundaries. 5. The caller receives authoritative result or a traceable error; support uses request/correlation evidence rather than secrets.

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Evidence status

Implemented on main. Evidence: services/auth-service/README.md; services/user-chef-service/README.md; services/catalog-service/README.md; services/order-service/README.md; services/subscription-service/README.md; services/integration-service/README.md; services/notification-service/README.md. Snapshot: main @ 3225f7fa531a (14 August 2026).

proof/media trust - end-to-end flow

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proof/media trust - errors and edge cases

Plain-English explanation

A useful failure model separates user input/state conflicts, identity/permission failures, dependency outages, data faults and deployment/configuration incidents.

How it works technically

order-service documents Problem Details fields and codes including ORDER_NOT_FOUND, ORDER_IDEMPOTENCY_CONFLICT, INVALID_STATUS, CANCELLATION_NOT_ALLOWED, REFUND_FAILED, UNAUTHORIZED, FORBIDDEN, BACKEND_CHANGE_DISABLED and BACKEND_UNAVAILABLE. Diagnostics should start from request/correlation evidence.

Flow described in words

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Data, security and deployment impact

The errors boundary owns its authoritative state; clients request or display changes but do not become the source of truth. Secrets and unnecessary personal data stay outside logs and PDFs. Releases must keep API, policy, schema and artifact compatible, with health/readiness evidence and rollback/backout appropriate to the change.

Evidence status

Implemented on main. Evidence: services/order-service/README.md; services/notification-service/README.md; apps/customer-web-next/README.md. Snapshot: main @ 3225f7fa531a (14 August 2026).

proof/media trust - deployment, security and operational validation

Plain-English explanation

The repository has broad CI/CD references across backend, frontend, mobile, APIM, data, privacy, security, smoke and release controls.

How it works technically

customer-web-next README describes ACR build/push, Container App revision deployment, readiness checks and restoration of the prior image/revision on failed readiness. Pipeline names not resolvable at their cited path are classified as reference-only instead of assigned invented triggers.

Flow described in words

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Data, security and deployment impact

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Evidence status

Implemented on main. Evidence: apps/customer-web-next/README.md; repository YAML search; docs/handover/. Snapshot: main @ 3225f7fa531a (14 August 2026).

order reconciliation - plain-language purpose

Plain-English explanation

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How it works technically

order-service documents idempotency, structured Problem Details errors, reconciliation, provider integration and controlled multi-database routing. Supported roles include customer, chef, admin and delivery-partner contexts with ownership enforcement.

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